

Ruokai Yin

Research Scientist, Meta

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RESEARCH INTERESTS

- Energy-efficient computer architectures, systems, and algorithm design for AI workloads, particularly those involving asymmetric, low-precision or sparse operands.

EDUCATION

Ph.D., Electrical and Computer Engineering, Yale University

Sep. 2021 — Apr. 2026

Thesis: Efficient Computing Architectures for Asymmetric AI Workloads

Advisor: Prof. Priyadarshini Panda

B.S., Electrical Engineering & Computer Science & Math, University of Wisconsin - Madison

Sep. 2018 — May. 2021

Graduated with Highest Distinction, GPA: 3.98/4.00

Advisor: Prof. Joshua San Miguel

EMPLOYMENT

AI Research Scientist, Reality Labs, Meta,

Apr. 2026 — present

Improving the runtime efficiency of AI models on Meta's custom silicon.

Research Intern, Azure-AI Architecture and System team, Microsoft,

May. 2025 — Aug. 2025

Mentors: Apala Guha & Xuan Zuo, Manager: Preyas Shah

RL-based autonomous parallelization strategy planning for distributed LLM inference systems at cloud server level.

Research Intern, ASIC team, Cerebras Systems,

May. 2024 — Aug. 2024

Mentor: Vipin Sharma

Architecture design and modeling for Cerebras's next-generation wafer-scale engine, with a focus on intra-PE, inter-PE, and IO level.

PUBLICATIONS [SELECTED]

Computer Architectures & Systems Design:

 **Learning to Shard: RL for Co-optimizing the Parallelism Degrees and Per-operator Sharding Dimensions in Distributed LLM Inference.**

[Ruokai Yin](#), Sattwik Deb Mishra, Xuan Zuo, Hokchhay Tann, Preyas Shah, Apala Guha

Workshop on ML for Systems at NeurIPS, 2025. **Spotlight Presentation.**

PacQ: A SIMT Microarchitecture for Efficient Dataflow in Hyper-asymmetric GEMMs.

[Ruokai Yin](#), Yuhang Li, and Priyadarshini Panda

ACM/IEEE Design Automation Conference (DAC) 2025.

LoAS: Fully Temporal-Parallel Dataflow for Dual-Sparse Spiking Neural Networks.

[Ruokai Yin](#), Youngeun Kim, Di Wu, and Priyadarshini Panda

International Symposium on Microarchitecture (MICRO) 2024.  [Open-source artifact.](#)

SATA: Sparsity-Aware Training Accelerator for Spiking Neural Networks.

[Ruokai Yin](#), Abhishek Moitra, Abhiroop Bhattacharjee, Youngeun Kim, and Priyadarshini Panda

IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2022.  [Open-source simulator.](#)

 **uGEMM: Unary Computing Architecture for GEMM Applications.**

Di Wu, Jingjie Li, [Ruokai Yin](#), Hsuan Hsiao, Younghyun Kim, Joshua San Miguel

International Symposium on Computer Architecture (ISCA) 2020, **IEEE Top-pick 2020.**  [Open-source simulator.](#)

Efficient AI Algorithm Design:

DuoGPT: Training-free Dual Sparsity through Activation-aware Pruning in LLMs.

[Ruokai Yin](#), Yuhang Li, Donghyun Lee, Priyadarshini Panda

Conference on Neural Information Processing Systems (NeurIPS), 2025.  [Open-source code.](#)

GPTAQ: Efficient Finetuning-Free Quantization for Asymmetric Calibration.

Yuhang Li, [Ruokai Yin](#), Donghyun Lee, Shiting Xiao, Priyadarshini Panda

International Conference on Machine Learning (ICML), 2025. [open-source code](#).

MINT: Multiplier-less Integer Quantization for Spiking Neural Networks.

Ruokai Yin, Yuhang Li, Abhishek Moitra, and Priyadarshini Panda

Asia and South Pacific Design Automation Conference (ASP-DAC) 2024, **Best Paper Award Nomination**. [open-source code](#).

Workload-balanced Pruning for Sparse Spiking Neural Networks.

Ruokai Yin, Youngeun Kim, Yuhang Li, Abhishek Moitra, Nitin Satpute, Anna Hambitzer, Priyadarshini Panda

IEEE Transactions on Emerging Topics in Computational Intelligence (TETCI), 2024. [open-source code](#).

Complete publication list available at [here](#).

AWARDS & HONORS

Research:

ML & Systems Rising Stars, MLCommons, 2026

Spotlight Paper, NeurIPS Workshop on ML for Systems, 2025

DAC Young Fellow, DAC, 2025

Best Paper Award Nomination, ASP-DAC, 2024

Spotlight Paper, NeurIPS Workshop on Learning from Time Series for Health, 2022

IEEE Micro Top Pick, Computer Architecture, 2020

Academic:

Conference Travel Fellowship, Yale University, Fall 2024

John Bennett Fenn Fellowship Fund, Yale University, Fall 2021 – Spring 2022

Distinctive Scholastic Achievement, University of Wisconsin - Madison, Spring 2021

Dean's Honor List, University of Wisconsin - Madison, Fall 2018 – Spring 2021

TALKS

PacQ: A SIMT Microarchitecture for Efficient Dataflow in Hyper-asymmetric GEMMs

62nd DAC (San Francisco, USA), June 2025

LoAS: Fully Temporal-Parallel Dataflow for Dual-Sparse Spiking Neural Networks

57th MICRO (Austin, USA), Nov 2024

MINT: Multiplier-less Integer Quantization for Energy Efficient Spiking Neural Networks

29th ASP-DAC (Incheon, South Korea), Jan 2024

SATA: Sparsity-Aware Training Accelerator for Spiking Neural Networks

Center for Brain-Inspired Computing (C-BRIC, SRC), Nov 2022

UnarySim and Characterizing Early Termination in Stochastic Computing

2020 UW Computer Architecture Industrial Affiliates (Madison, WI, USA), Sep 2020

TEACHING EXPERIENCE

TA - EENG 439, Neural Networks & Learning Systems, Fall 2023

Instructor: Prof. Priya Panda

TA - EENG 348, Digital Systems, Spring 2023

Instructor: Prof. Rajit Manohar

ACADEMIC ACTIVITIES

Reviewer

- IEEE Transactions on Neural Networks and Learning Systems
- IEEE International Symposium on Circuits and Systems, 2024
- IEEE Journal on Emerging and Selected Topics in Circuits and Systems
- IEEE Transactions on Very Large Scale Integration Systems
- IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems
- Frontiers in Neuroscience

- Pattern Recognition
- Neural Networks
- Expert Systems With Applications